

LUCY ISLAND LIGHTSTATION, BC, Canada

WMO: 712200

Lat: 54.296N

Lon: 130.609W

Elev: 23

StdP: 101.05

Time Zone: -8.00 (NAP)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) -5.0	(c) -3.1	(d) -16.7	(e) 0.9	(f) -3.3	(g) -12.1	(h) 1.3	(i) -0.6	(j) 23.5	(k) 6.2	(l) 22.0	(m) 5.7	(n) 5.5	(o) 50	(p) 0.783

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	3.4	17.7	14.6	16.6	14.3	15.7	13.8	15.5	16.8	14.9	16.1	14.3	15.5	3.5	270

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
14.8	10.6	15.9	14.2	10.1	15.4	13.7	9.8	15.0	43.4	16.8	41.7	16.2	40.2	15.6	17.4

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a) 20.2	(b) 18.0	(c) 15.7	(e) -7.2	(f) 21.1	(g) 3.4	(h) 2.4	(i) -9.7	(j) 22.8	(k) -11.7	(l) 24.2	(m) -13.6	(n) 25.6	(o) -16.1	(p) 27.3		
			DB													
			WB													

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	8.2	3.8	3.9	4.7	6.9	9.3	11.5	13.2	13.7	12.1	9.1	6.0	4.3		
	DBStd	4.28	3.36	2.80	2.64	2.00	2.01	1.76	1.58	1.44	1.64	1.90	2.80	2.81		
	HDD10.0	1001	192	171	165	96	36	4	1	0	2	38	121	176		
	HDD18.3	3682	450	404	423	343	279	206	158	143	187	286	369	434		
	CDD10.0	362	0	0	0	2	16	49	101	116	66	11	1	0		
	CDD18.3	0	0	0	0	0	0	0	0	0	0	0	0	0		
	CDH23.3	0	0	0	0	0	0	0	0	0	0	0	0	0		
	CDH26.7	0	0	0	0	0	0	0	0	0	0	0	0	0		
(14) Wind	WSAvg	6.3	7.6	7.1	7.2	6.3	5.3	4.8	4.3	4.4	5.5	7.0	7.8	8.1		
(15) Precipitation	PrecAvg	2364	243	180	190	158	101	98	104	155	216	309	291	270		
	PrecMax	3048	509	312	401	310	282	172	189	297	471	466	527	575		
	PrecMin	1641	90	96	64	87	9	31	50	64	93	108	143	109		
	PrecStd	380	85	64	72	56	63	35	38	70	95	100	86	100		
(20) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	10.1	10.2	11.4	14.9	17.4	19.3	18.4	19.6	18.0	14.4	12.0	11.0		
		MCWB	8.1	8.0	8.1	9.3	11.1	15.2	15.3	15.9	14.6	11.8	9.6	7.7		
	2%	DB	9.0	9.1	9.6	12.1	14.3	16.2	17.0	17.6	16.1	13.1	10.7	9.7		
		MCWB	7.5	7.3	7.2	8.8	11.1	13.6	14.8	15.0	13.3	11.2	9.4	7.5		
	5%	DB	8.2	8.2	8.5	10.3	12.8	14.7	16.0	16.6	15.0	12.2	10.1	8.8		
		MCWB	6.8	6.6	6.6	7.9	10.5	12.8	14.2	14.6	13.3	10.4	9.1	6.9		
	10%	DB	7.5	7.3	7.8	9.3	11.8	13.7	15.3	15.7	14.2	11.5	9.4	7.9		
		MCWB	6.2	5.9	6.1	7.3	9.9	12.2	13.8	14.0	12.8	9.9	8.1	6.3		
(28) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	8.8	9.2	9.5	10.2	12.2	15.1	16.0	16.4	15.5	13.2	10.7	8.6		
		MCDB	9.3	9.9	10.9	11.9	15.3	17.3	17.3	18.1	17.0	13.8	11.4	10.2		
	2%	WB	7.9	8.0	8.1	9.2	11.3	14.0	15.4	15.6	14.5	12.3	10.0	7.7		
		MCDB	8.7	9.0	9.5	10.6	13.2	15.6	16.6	16.8	15.8	13.0	10.7	9.2		
	5%	WB	7.0	7.2	7.4	8.3	10.8	13.3	14.8	15.0	13.9	11.1	9.5	7.1		
		MCDB	8.0	8.3	8.9	9.7	12.3	14.8	16.0	16.1	15.1	12.0	10.3	8.5		
	10%	WB	6.3	6.4	6.7	7.7	10.2	12.8	14.3	14.4	13.3	10.4	8.7	6.4		
		MCDB	7.3	7.6	8.2	9.1	11.5	14.1	15.4	15.4	14.3	11.4	9.7	7.7		
(35) Mean Daily Temperature Range	5% DB	MDBR	2.9	3.1	3.4	3.6	3.5	3.3	3.2	3.4	3.3	3.0	3.0	2.9		
		MCDBR	3.3	3.6	4.0	4.9	4.9	4.8	4.3	4.8	4.4	3.6	3.4	3.7		
	5% WB	MCWBR	2.6	2.5	2.6	2.6	2.7	2.5	2.3	2.6	2.4	2.4	2.6	2.4		
		MCWBR	3.1	3.3	3.7	3.9	4.2	4.0	3.6	4.4	4.4	3.3	3.0	3.4		
(40) Clear-Sky Solar Irradiance	taub		0.304	0.311	0.336	0.371	0.363	0.377	0.368	0.363	0.355	0.337	0.312	0.294		
	taud		2.393	2.506	2.451	2.351	2.440	2.422	2.493	2.527	2.521	2.483	2.366	2.325		
	Ebn at Noon		658	797	847	856	883	871	874	859	818	747	631	578		
	Edh at Noon		54	67	86	109	105	108	99	91	81	67	54	46		
(44) All-Sky Solar Radiation	RadAvg		0.51	1.33	2.39	3.82	4.90	5.17	4.68	4.04	2.67	1.47	0.68	0.41		
	RadStd		0.07	0.23	0.29	0.23	0.45	0.39	0.53	0.36	0.30	0.21	0.09	0.07		

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	N/A	+0.74	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (18 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page